Take-home Assignment – Agentic Customer-Service Assistant

Role: AI / LLM Engineer · Suggested time: 5 days · Materials: *materials/* folder

1. Context

Lumen Home (a fictional company) sells air conditioners, washing machines, refrigerators and water purifiers across Vietnam. Its customer-service (CS) team answers thousands of chat and phone questions a day about prices, promotions, warranty, returns, installation fees and service centres. The knowledge lives in a few spreadsheets maintained by different departments and a policy handbook in PDF. Your job is to build an assistant that answers these questions **correctly, fast and only from the provided knowledge**.

2. Materials provided

File	Content
product_catalog.xlsx	Several sheets: products, regional price list, promotions, warranty terms, province → price region.
service_network.xlsx	Service centres (address, hours, capabilities, status) and the technical-service fee schedule.
policy_handbook.pdf	CS policy handbook v3.2: returns, warranty, installation, compensation, escalation.
sample_qa.xlsx	8 example customer questions with the time they were asked and a reference answer.

The files were made by people, not for machines: expect merged and multi-row headers, notes and footnotes inside the sheet, several valid-from dates, rules split between files, and information that is outdated or that contradicts another file. Reading the materials carefully is part of the task. Questions and answers are in Vietnamese.

3. What to build

A service that takes a customer question (plus the time it was asked, and the earlier turns of the conversation when there are any) and returns an answer to the customer. It must:

- **Answer correctly** using only the materials, including questions that need several facts or a small calculation.
- **Cite its sources** (file + sheet/row or handbook section) for every fact the answer relies on.
- **Respect time**: prices, promotions and centre status depend on when the question is asked.
- **Know its limits**: ask a short follow-up question when key information is missing; say clearly when something is out of scope or not in the materials. Never invent prices, policies, addresses or phone numbers.
- **Be agentic**: the system decides for itself what to look up and how to combine it, rather than following a fixed script per question type.
- **Stay maintainable**: CS staff update the spreadsheets and the handbook every week. A new version must be usable without code changes.

4. Latency and cost targets

Metric (measured end-to-end, on your setup)	Target
Full answer, p50 / p95	≤ 3 s / ≤ 7 s
Time to first token, if you stream	≤ 1.5 s (p50)
Cost per question	Report it; justify your model choices

A fast answer that is wrong is worth less than a slightly slower correct one; we want to see how you balance the two and what you measured.

5. Interface

Expose at least `POST /ask` with input `{"question", "asked_at", "history"?}` and output `{"answer", "citations", "latency_ms"}`. A minimal chat UI is optional. Any language, framework, LLM provider or

open model is allowed; state which ones you use. Do not hard-code answers to the sample questions – we grade on a **hidden set of similar questions**, and will also run the system against updated versions of the materials.

6. Deliverables

1. **Code repository** with a README: one command to set up, one to run, and how to point it to a new version of the materials.
2. **Design document** (max 4 pages): architecture, how the materials are processed, how the agent decides what to do, how you keep answers grounded and fast, failure modes you know about, and what you would do next with more time.
3. **Evaluation report**: your evaluation method, results on the sample set plus at least 15 questions you wrote yourself (including hard and adversarial ones), latency distribution and cost. Explain how you judged correctness.
4. **Demo**: a 5-minute recording or a live session.

7. How we evaluate

Criterion	Weight
Answer correctness on the hidden set (facts, numbers, dates, follow-up questions, refusals)	35%
Grounding: correct citations, no fabricated information	15%
Latency and cost against the targets, and how honestly they are measured	15%
Design quality and trade-off reasoning (design doc + interview)	20%
Code quality, reproducibility, handling of updated materials	10%
Evaluation method and self-written test set	5%

Questions about the brief: send them to your recruiter contact. If something is ambiguous, make a reasonable assumption and write it down in the design document.